

Instruction to the candidates:

1. **Special credit will be given for answer which are brief and to the point.**
2. **Figure in the margin indicates full marks for the question.**

PART – B (Marks-10)

1. Find out the correct answer out of the option given against each question:

(1X10=10)

- (i) A coin is tossed 10 times. The probability of getting head 6 times:

(a) ${}^{10}C_5 \times \frac{1}{2^{10}}$

(b) ${}^{10}C_3 \times \frac{1}{2^{10}}$

(c) ${}^{10}C_4 \times \frac{1}{2^{10}}$

(d) ${}^{10}C_8 \times \frac{1}{2^{10}}$

- (ii) The rate of increase of a side of a square is 1 cm/sec. The rate of increase of area of the square, when length of a side of the square is 2 cm, is:

(a) $4 \text{ cm}^2/\text{sec}$

(b) $8 \text{ cm}^2/\text{sec}$

(c) $1 \text{ cm}^2/\text{sec}$

(d) $2 \text{ cm}^2/\text{sec}$

- (iii) The angle between the planes $x - y + 2z = 9$ and $2x + y + z = 7$ is:

(a) 30°

(b) 45°

(c) 60°

(d) 90°

- (iv) If $P(A) = \frac{3}{7}$, $P(B) = \frac{4}{7}$ and $P(A \cap B) = \frac{2}{9}$, then find the value of $P(A/B)$:

(a) $\frac{7}{18}$

(b) $\frac{14}{27}$

(c) $\frac{5}{18}$

(d) $\frac{4}{9}$

- (v) If two rows or two columns of a determinant are identical then value of the determinant is

(a) 0

(b) 2

(c) -1

(d) 1

- (vi) The value of $\tan\left(\frac{\pi}{2} - \tan^{-1}\frac{1}{3}\right)$ is equal to:

(a) $\frac{1}{3}$

(b) 3

(c) $\frac{2}{3}$

(d) $\frac{3}{2}$

- (vii) If $f(x) = -f(-x)$, then the value of $\int_{-a}^a f(x)dx$ is equal to:

(a) 2a

(b) a

- (c) $\frac{a}{2}$ is equal to (d) 0
- (viii) If two vectors $\vec{a} = \hat{i} + 3\hat{j} - \hat{k}$ and $\vec{b} = 2\hat{i} + 6\hat{j} + \lambda\hat{k}$ are parallel then the value of λ is:
- (a) 3 (b) -6
(c) -3 (d) -2
- (ix) The domain for which the function $f(x) = 3x^2 - 2x$ and $g(x) = 3(3x - 2)$ are equal, will be:
- (a) $\left\{1, \frac{2}{3}\right\}$ (b) $\{1, 3\}$
(c) $\left\{\frac{2}{3}, 3\right\}$ (d) $\left\{\frac{2}{3}, 0\right\}$
- (x) If $\tan^{-1} \frac{5-x}{1+5x}$, then value of $\frac{dy}{dx}$ is :
- (a) $-\frac{1}{1+x^2}$ (b) $\frac{1}{1+x^2}$
(c) 5 (d) $\frac{5}{1+x^2}$

PART – A (Marks : 70)

1. Answer the following questions (alternatives are to be noted)

- (a) Answer any one: (2X1=2)
- (i) Solve: $2 \sin^{-1} x = \cos^{-1} x$, $0 < x < 1$.
- (ii) Let $A = \{1, 2, 3\}$. Define a relation on A which is reflexive and symmetric but not transitive.
- (b) Answer any one: (2X1=2)
- (i) $A = \begin{bmatrix} 8 & 0 \\ 4 & -2 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & -2 \\ -5 & 1 \end{bmatrix}$, find another matrix X where $2A = 3X = 5B$.
- (ii) If $\begin{vmatrix} 2 & 3 \\ 4 & 5 \end{vmatrix} = \begin{vmatrix} x & 3 \\ 2x & 5 \end{vmatrix}$, find the value of x.
- (c) Answer any three questions: (2X3=6)
- (i) If $y = \sin^{-1} \frac{2x}{1+x^2}$, find $\frac{dy}{dx}$.
- (ii) $f(x) = 5 - |x - 1|$, find the maximum value of f(x), also find the value of x for which f(x) maximum.
- (iii) If $x > 0$, then show that $\log(1+x) > \frac{x}{1+x}$.

- (iv) Find the differential equation of the circles which touch the x axis at origin.
- (v) If $f(2) = 4$, $f'(2) = 4$, then evaluate $\lim_{x \rightarrow 2} \frac{xf(2) - 2f(x)}{x - 2}$.
- (vi) Evaluate $\int_1^2 \frac{x dx}{(x+1)(x+2)}$.
- (d) Answer any one: (2X1=2)
- (i) Show that A(2, 3, -4), B(1, -2, 3) and C(3, 8, -11) are collinear.
- (ii) If $\vec{a} = 5\hat{i} - \hat{j} - 3\hat{k}$ and $\vec{b} = \hat{i} + 3\hat{j} - 5\hat{k}$, then show that $\vec{a} + \vec{b}$ and $\vec{a} - \vec{b}$ are perpendicular to each other.
- (e) Answer any one: (2X1=2)
- (i) If $P(A) = a$ and $P(B) = b$, then show that $P(A/B) \leq \frac{a}{b}$.
- (ii) If a and b are constants, then show that $\text{var}(ax + b) = a^2 \text{var}(x)$.

2.

- (a) Answer any one: (4X1=4)
- (i) R_1 and R_2 are two equivalence relation defined on Set $A (\neq \phi)$. Show that $R_1 \cap R_2$ is an equivalence relation.
- (ii) If $\tan^{-1} x + \tan^{-1} y + \tan^{-1} z = \frac{\pi}{2}$ and $x + y + z = \sqrt{3}$, then show that $x = y = z$.
- (b) Answer the following questions: (4X2=8)
- (i) If $A = \begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{pmatrix}$, then show $A^2 - 4A - 5I_3 = 0$.
- OR
- If $A = \begin{pmatrix} i & -i \\ -i & i \end{pmatrix}$ and $B = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$, then show that $A^8 = 128B$.
- (ii) Show that $\begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 1+c \end{vmatrix} = abc \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right)$, $abc \neq 0$.
- OR
- Using Cramer's rule solve the following equations:
 $3x + y + z = 10$, $x + y - z = 0$, $5x - 9y = 1$
- (c) Answer the following Questions: (4X3=12)
- (i) If $\cos y = x \cos(a + y)$, $a \neq 0$, then show that $\frac{dy}{dx} = \frac{\cos^2(a + y)}{\sin a}$.
- OR

If $x = \sin t$, $y = \sin kt$, $k \neq 0$, then show that

$$(1 - x^2) \frac{d^2 y}{dx^2} - x \frac{dy}{dx} + k^2 y = 0.$$

(ii) Evaluate: $\int \sqrt{1 - \sec x} dx$

OR

$$\text{Evaluate: } \int \frac{dx}{(x-1)\sqrt{x^2-1}}$$

(iii) Solve: $(e^x + 1)ydy - (y^2 + 1)e^x dx = 0$, given $x = 0$, $y = 0$.

OR

$$\text{Solve: } (x dy - y dx) y \sin \frac{y}{x} = (y dx + x dy) x \cos \frac{y}{x}.$$

(d) Answer any one: (4X1=4)

(i) $\vec{a}, \vec{b}, \vec{c}$ be three vectors such that $\vec{a} + \vec{b} + \vec{c} = \vec{0}$ and $|\vec{a}| = 1, |\vec{b}| = 4, |\vec{c}| = 2$, evaluate $\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}$.

(ii) If sum of two unit vectors be a unit vectors, then show that difference of those two vectors is $\sqrt{3}$.

(e) Answer any one: (4X1=4)

(i) Using integral calculus, find the area bounded by $\frac{x^2}{2} + \frac{y^2}{1} = 1$.

(ii) Prove that $\int_1^3 \frac{dx}{x^2(x+1)} = \frac{2}{3} + \log \frac{2}{3}$.

(f) Answer any one: (4X1=4)

(i) $A_1, A_2, A_3, \dots, A_n$ are independent and $P(A_i) = 1 - q_i$ ($i = 1, 2, \dots, n$). Show that $P(A_1 \cup A_2 \cup A_3 \cup \dots \cup A_n) = 1 - q_1 q_2 q_3 \dots q_n$.

(ii) Eight unbiased coins tossed simultaneously. Find the probability of getting exactly five heads and at least five heads.

3.

(a) Answer any one: (5X1=5)

(i) A manufacturer produces two models A and B of a product. Each piece of model A requires 9 labour hours for fabricating and 1 labour hour for finishing. Each piece of product B requires 12 labour hour for fabricating and 3 labour hour for finishing. For fabricating and finishing the maximum labour hour available are 180 and 30 respectively. The company makes a profit of Rs. 8000 on each piece of model A and Rs. 12000 on each piece of model B. Formulate an LPP so as to maximize his profit.

(ii) Solve the following LPP (graph sheet not required)

Minimize $Z = 200x + 500y$

Subject to the constraints: $x + 2y \geq 10$

$$3x + 4y \leq 24$$

$$x, y \geq 0$$

(b) Answer any two: (5X2=10)

(i) $ax^2 + by^2 = 1$, and $Ax^2 + By^2 = 1$ meet each other orthogonally

($a \neq A, b \neq B, aB - bA \neq 0$). Show that $\frac{1}{a} - \frac{1}{b} = \frac{1}{A} - \frac{1}{B}$.

(ii) Solve $(1 + x^2)dy + 2xydx = \cot x dx$

(iii) Evaluate: $\lim_{n \rightarrow \infty} \left(\frac{n}{n^2 + 1^2} + \frac{n}{n^2 + 2^2} + \dots + \frac{1}{2n} \right)$

(c) Answer any one: (5X1=5)

(i) Find the equation of the plane which passes through $(-2, 1, 3)$ and through the intersection of the planes $2x - 7y + 4z = 0$ and $3x - 57 + 4z + 11 = 0$.

(ii) Find the point on the line $\frac{x+2}{3} = \frac{y+1}{2} = \frac{z-3}{2}$ at a distance $3\sqrt{2}$ units from the point $(1, 2, 3)$.
